JAIMINA GHARIA

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🌐 Jaimina Gharia | LinkedIn

BIOTECHNOLOGY STUDENT

Biotechnology undergraduate with extensive research experience spanning molecular biology, metabolomics, and computational modeling. Proficient in mammalian cell culture, as well as molecular and cellular biology techniques. Author of peer-reviewed oncology review article and eager to leverage strong foundations in cellular mechanisms and analytical techniques to pursue a neuroscience dissertation focusing on neurobiology, cellular signaling, or neuro-degeneration.

EDUCATION

GSFC University, Vadodara
Bachelor of Science (Hons.) in Biotechnology
Current CGPA: 7.94/10

PUBLICATIONS

Revolutionizing Cancer Care: Bioprinting Prostate Cancer Stem Cells for Targeted Treatments
World Journal of Clinical Oncology
DOI | PubMed

EXPERIENCE

Research Intern | Charotar University of Science and Technology (CHARUSAT) **June 2026**

- **HPLC:** Operated HPLC for the separation and quantification of compounds from mushroom extract.
- **LC-MS & LC-MS/MS:** Utilized advanced LC-MS and LC-MS/MS instrumentation to perform structural characterization and qualitative analysis of mushroom samples.
- **Bioinformatics:** Analyzed mass spectrometry data to identify unknown metabolites/impurities using R.

Research Intern | Animal Tissue Culture Lab, GSFC University **June 2025 – May 2026**

- **Mammalian Cell Culture:** Successfully maintained different cell line cultures and conducted cell viability assays including MTT and Clonogenic assay.
- **Fluorescence Microscopy:** Utilized phase-contrast and fluorescence microscopy to conduct functional assays (scratch, MTT), tracking real-time cellular migration and proliferation dynamics.
- **Dosing:** Gained proficiency in dosing experiments and successfully found distinct effects of variable concentrations of the same drug.

Research Intern | Biochemistry Department, MSU Baroda **December 2025**

- **Western Blotting:** Executed western blotting workflows for targeted protein separation and analysis.
- **Fluorescence Microscopy:** Executed JC-1 and TMRM staining assays utilizing fluorescence microscopy to evaluate mitochondrial membrane potential.
- **Molecular Techniques:** Performed total RNA isolation, cDNA synthesis, and downstream PCR amplification.

Research Intern | Chronobiology Lab, Zoology Department, MSU Baroda **December 2024 - February 2025**

- **Molecular Biology:** Mastered and applied molecular biology techniques including plasmid isolation, microbial culture, transfection, and transformation.
- **Cell culture and Animal Handling:** Assisted with cell culture and basic animal handling under supervision gaining insight into behavioral/physiological rhythms.

EXPERIENCE (CONT.)

Intern | ETP Laboratory, GSFC Ltd. **June 2024 - July 2024**

- **Analytical Techniques:** Analyzed effluent for key environmental parameters, including BOD, COD, and suspended solids (SS).
- **Laboratory Exposure:** Gained hands-on experience in routine laboratory testing and sample analysis.

CONFERENCE PRESENTATIONS

Evaluating the Effect of AR Antagonists on Cellular Proliferation in Prostate Cancer **December 20, 2025**
Presented at Charotar University of Science and Technology (CHARUSAT), Anand

Comparative Insights into CRISPR-Cas9, Cas12, and Cas13 in Cancer Therapy and Addressing Off-target Challenges **April 10, 2025**
Presented at Pandit Deendayal Energy University (PDEU), Gandhinagar

Advances in Personalized Medicine for HPV-related Cancers **March 4, 2025**
Presented at HPV & Cancer: Exploring Advances in Prevention and Care Symposium, Vadodara

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PROJECTS

Scratch Assay Analysis Software (Collaborative Project)

- Co-developing a software tool for quantifying results of wound healing assays, improving reproducibility and data analysis accuracy.

Metagenomics Analysis of Biogas Plant Samples

- Analyzed metagenomic sequencing data and co-preparing a manuscript detailing microbial diversity in biogas systems.

Plant Growth-Promoting Microbial Isolates (Pot Experiment)

- Designed and conducted pot experiments to evaluate the growth-promoting potential of selected microbial isolates in plants.

Shoot Induction and Regeneration in Eggplant

- Optimized hormone treatments for seed-based tissue culture to induce shoot formation and regenerate whole plants.

ACHIEVEMENTS

Student of the Year - GSFC University **2025-2026**

- Conferred in recognition of outstanding academic excellence, consistent curricular and co-curricular achievements, and exemplary discipline.

STRENGTHS AND EXPERTISE

Technical Skills:
Wet-Lab Skills:
Mammalian Cell Culture | HPLC | LC-MS & LC-MS/MS | Western Blotting | DNA/RNA Isolation |
cDNA Synthesis & PCR | Transfection & Transformation | Fluorescence & Phase-Contrast Microscopy |
Microbial Culture

Computational Skills:
Data interpretation | Python & R (Bioinformatics) for image analysis | LC-MS data analysis

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